Eigen-Séta v2 L3/L5 round-trip record

Run `run-20261011T1946Z-l3-l5-fix`, protocol frozen 2026-10-11 19:46 UTC. Source: repository commit `cd47963f9` imported into base `7f7672d01`, with the local constant-time scalar-recovery edge-case correction. The runs do not cover the subsequent public-input validation edit.

Commands from `eigen_seta/`, run concurrently: `cargo run --release --example run -- L3 1` and `cargo run --release --example run -- L5 1`. Rust 1.99.0, release build; AMD EPYC 7763 shared cloud VM, x86-64, five logical CPUs. No core pinning, isolation or frequency control. `OsRng` seeds not captured.

Set	KeyGen (s)	Encaps (s)	Decaps (s)	Correct
L3	112.612	1.3059	47.275	1/1
L5	259.034	1.9862	107.221	1/1

Requested and completed: one key and one encapsulation–decapsulation per set. Both processes exited 0; no failures. Each time is one measured wall-clock sample (n=1); no spread or uncertainty interval was recorded. Public key and ciphertext were 1992 bytes for L3 and 2632 bytes for L5. Sampled secret-degree bit lengths: 409 and 536. Exact output: `records/run_output.txt`.

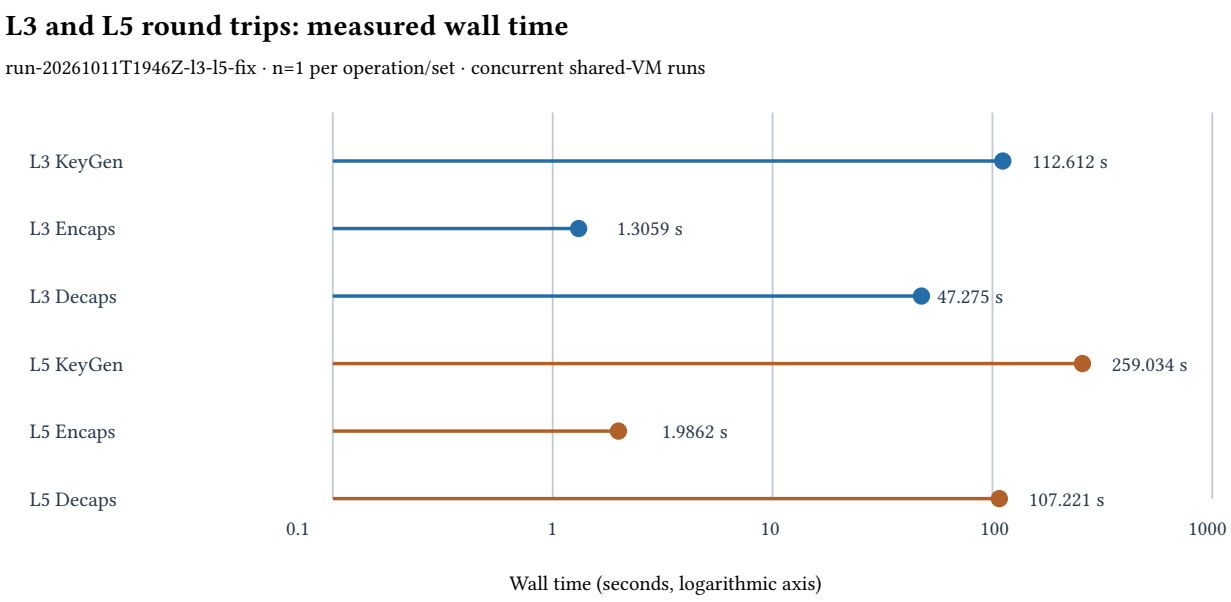

Figure 1: Wall times, one sample per operation and set, log seconds. Run run-20261011T1946Z-l3-l5-fix.

This demonstrates functional round trips. Concurrent execution on a shared VM prevents an isolated performance comparison. These samples do not establish security or constant-time behavior.